

SHASHANK SINGH GAUTAM

Python Full Stack Developer

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PROFESSIONAL SUMMARY

Full Stack Developer with 4+ years of experience at CGI India delivering enterprise systems using **Django, React.js, FastAPI, and LangChain**. Specializes in backend architecture, Agentic AI pipelines, event-driven microservices, and cross-platform data synchronization. Delivered measurable impact across 4 production platforms - including **90% reduction in manual processing** and **70% faster incident response** - recognized with 3 CGI performance awards.

TECHNICAL SKILLS

Languages : Python, Java, JavaScript
Backend : Django, FastAPI, REST APIs, Microservices, Multi-threading, Pandas
Frontend : React.js, Redux Toolkit, Tailwind CSS, Bootstrap, HTML5, Vite
AI / Cloud : LangChain, LangGraph, Google ADK, Azure OpenAI (GPT-4o), Agentic AI, AWS
Messaging : RabbitMQ, Event-Driven Architecture
Testing : Selenium, Postman, Unit Testing, API Testing, Swagger
Databases : PostgreSQL, MongoDB, NeonDB, Elastic Search
DevOps : Docker, Linux, Nginx, Gunicorn, Git, Github
Concepts : OOP, SOLID, RAG, DSA, CI/CD, Agile, Database Design, Query Optimization

PROFESSIONAL EXPERIENCE

CGI India - BELL Automation Team
Full Stack Developer

June 2022 – Present

Agentic AI Operations Platform (AIOPS)

- Built an **agentic reasoning loop** using **LangChain + GPT-4o** with up to 15-step iterative tool invocation, dynamic REST API execution, and contextual response generation.
- Developed a **FastAPI** backend with async tool execution, **RabbitMQ** distributed logging, and real-time activity streaming to a React chat UI.
- Designed a **multi-provider LLM service** supporting **Azure OpenAI, Ollama, and LM Studio** with singleton caching, reducing initialization overhead by **80%** for concurrent requests.

Shift Roster Management System

- Architected a **normalized, microservice-oriented roster system** supporting team setup, roster cloning, automated email distribution, and high-volume schedule processing.
- Built an **interactive scheduling module** using **React.js and Django REST Framework** with on-call linking, roster exports, and SDM-level role-based access control.
- Integrated **Twilio voice conferencing**, leave management, and notifications to reduce manual coordination during critical shift handoffs.
- Optimized concurrent task handling via **ThreadPoolExecutor, asyncio, and Django signals**, improving overall platform responsiveness.

Adhoc Opportunity Management Platform

- Developed a **full-stack event-driven workflow platform** for SDMs, Directors, and Clients to onboard and track adhoc opportunities with real-time funding and approval visibility.
- Built **role-based dashboards and KPI analytics** with quarter/month/year-wise reporting to support leadership-level operational tracking.
- Engineered a **multi-level email approval chaining engine** with dynamic stakeholder selection and async event-driven notifications.
- Containerized the platform using **Docker** and deployed demo environments across multiple enterprise clients for onboarding.

Generic Pagination & Dynamic API Framework

- Engineered a reusable **generic paginated API framework** in **Django REST Framework** supporting dynamic searching, sorting, advanced filtering, and customizable query handling across multiple enterprise modules.
- Developed a **custom pagination engine** with configurable page size, metadata-driven responses, and **full dataset export support** using dynamic `download="all"` handling for large-scale reporting workflows.
- Built extensible **class-based API views** with serializer abstraction, reusable filter utilities, dynamic ordering, and centralized response formatting to reduce duplicate backend development effort.
- Implemented optimized **queryset filtering, distinct column metadata generation, and dynamic search pipelines**, improving frontend table rendering and enterprise dashboard usability.

Subscription & Notification Engine

- Built an **event-driven pub-sub notification engine** for real-time ITSM incident alerts, reducing response times by **70%**.
- Refactored **database schema and Django ORM query layer**, resolving normalization bottlenecks and improving query performance by **60%**.
- Implemented **RBAC, crontab scheduling, and Python lru.cache optimization** to improve alert throughput and reduce redundant computations.

Application 360°

- Architected a **microservices ecosystem** using **Django, React.js, and REST APIs** achieving seamless SmartOps–SmartHub synchronization with a **99% transaction success rate**.
- Automated **cross-platform JSON sync pipelines** triggered via email between SmartOps (India) and SmartHub (Canada), improving processing efficiency by **90%**.
- Designed a **centralized logging and audit framework** with anomaly detection, retry mechanisms, and daily compliance reporting for enterprise-grade visibility.
- Implemented **JWT authentication and Django class-based API authorization** to enforce secure role-based access across enterprise modules.

EDUCATION

Indian Institute of Information Technology, Senapati

B.Tech in Electronics and Communications Engineering CGPA: 7.66

Imphal, Manipur, India

Aug 2018 – May 2022

AWARDS & INTERESTS

- Received **1 Silver (2026) and 2 Bronze Awards (2024, 2025)** at CGI for full-stack development and automation contributions - [View Certificate](#).
- Interests: System Design, Backend Engineering, Distributed Systems, Table Tennis, Cricket, and Cooking.